

NEW PRODUCTS

TECHNOLOGY

COMPONENTS

Apollo330 Plus SoCs

Ambiq's Apollo330 Plus SoC series enhances AI edge computing with ultra-low power consumption. It includes Apollo330 Plus, Apollo330B Plus, and Apollo330M Plus, designed for healthcare, smart homes, and industrial applications. Powered by the Cortex-M55 with Helium technology, it accelerates AI processing with 16x speed and 30x better energy efficiency. Supporting Bluetooth low energy, matter, and thread, it extends device lifetimes while enabling seamless connectivity and real-time AI inferencing.

Ambiq
<https://ambiq.com>



GaN transistors

Infineon Technologies AG introduces the CoolGaN G3 Transistor 100V (IGD015S10S1) in RQFN 5 × 6 and 80V (IGE033S08S1) in RQFN 3.3 × 3.3, setting a new industry standard. These transistors offer seamless silicon integration, enhanced efficiency, and multi-sourcing options. Their advanced packaging ensures low resistance, optimised heat dissipation, and superior thermal performance. Ideal for datacom, consumer, and industrial power applications, they simplify GaN adoption.

Infineon Technologies AG
<https://www.infineon.com/cms/en>



BM-ICs for 48V batteries

Nuvoton Technology Corporation Japan (NTCJ) has developed the

KA49701A and KA49702A 17-cell BM-ICs for 48V batteries, enhancing battery system safety and simplifying system construction. These ICs



integrate fault diagnosis and fail-safe functions, eliminating external protection circuits and reducing costs. Lower power consumption extends battery life and minimises self-discharge, supporting long-term storage. Designed for energy storage and AI data centres, they improve lithium-ion battery safety while reducing costs in high-capacity systems.

Nuvoton Technology Corporation
<https://www.nuvoton.com>

Ultra-low-power MCUs

Renesas Electronics has launched the RA4L1 MCU group with 14 ultra-low-power devices, ideal for water meters, smart locks, and IoT sensors. Consuming just 168µA/MHz in active mode and 1.70µA in standby, they suit portable printers, digital cameras, and smart labels. The flexible software package (FSP) simplifies development, supporting RTOS, middleware, and seamless migration across Renesas RA-series MCUs.



Renesas Electronics Corporation
<https://www.renesas.com/en>

Protective diodes for ECUs

ROHM has introduced the ESDCANxx series of bidirectional TVS diodes for high-speed in-vehicle communication using



CAN FD. These diodes protect ECUs from surges and ESD while maintaining signal integrity. The 27V DFN1010 models offer 3.2x higher surge rating and 16% lower clamping voltage than standard products.

ROHM Semiconductor
<https://www.rohm.com>

Hollow shaft ring encoders

Same Sky's Motion and Control Group has introduced hollow shaft ring encoders to its panel mount incremental encoder lineup. Available in sizes from 23mm to 56.5mm, these low-profile encoders suit industrial, commercial, and home appliance applications. With hollow shaft diameters from 9mm to 38.5mm and precise motion feedback, they enhance automation, robotics, and machinery.

Same Sky
<https://www.sameskydevices.com>



NTC thermistors

TDK Corporation has launched two new NTC thermistor series: the L862 (B57862L) with bendable wires and the L871 (B57871L) with fixed lead spacing, designed for automotive, industrial, and battery system applications. With rated resistances from 1kΩ to 100kΩ, they ensure high accuracy and fast response. Ideal for energy storage, drones, and temperature-sensitive systems, they provide reliable thermal monitoring solutions.

TDK Corporation
<https://www.tdk.com/en/index.html>



All-band GNSS module

The u-blox ZED-X20P is an all-band GNSS module delivering centimetre-level accuracy at a fraction of